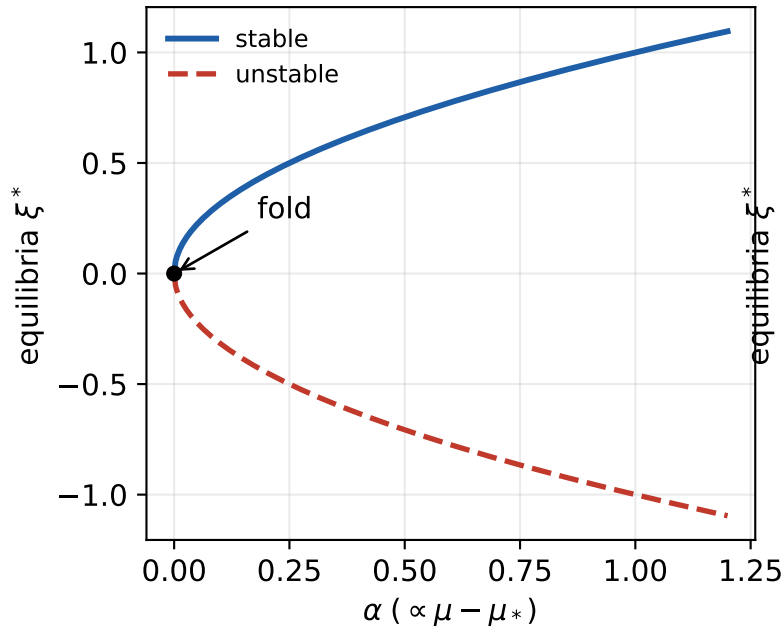


Codimension-1 bifurcations: the cubic comes from $\det \Gamma$

(a) Fold (saddle-node) $\dot{\xi} = \alpha - \xi^2$



(b) Pitchfork $\dot{\xi} = \lambda \xi - \xi^3$

